Portfolio

\# AI Scientist

\# Efficient Vision Generative Model

\# Multimodal Empathy Model

이아현 Ahhyeon Lee
M.S. in IT Engineering, Sookmyung Women's University (Seoul, Republic of Korea)

ahlee.sep@gmail.com | ah.lee@ivpl.sm.ac.kr
(+82) 010-9109-6271
https://github.com/ahlee-smwu

Contents

Master's Thesis (1/3)

Topic: Few-Step Diffusion via Adaptive Prior

Title: RAPID: Robust Adaptive Prior Integration Diffusion

Research Objectives

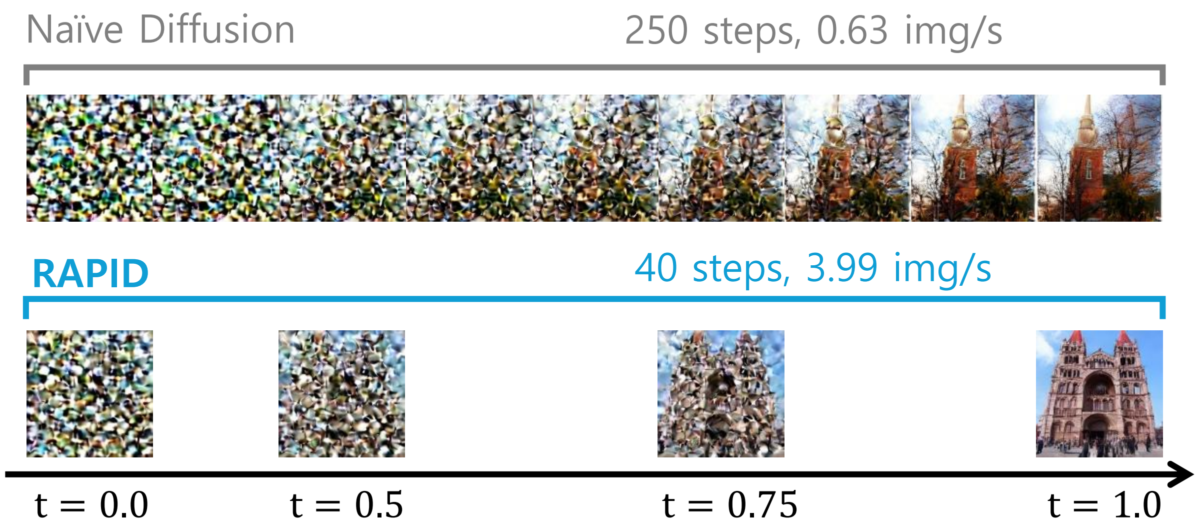
- ➢ Few-Step generating Diffusion
- ➢ Data and step adaptive prior in Diffusion

Motivation

- ➢ Problem Definition
  - Too slow sampling in Diffusion.
  - Diffusion's capacity is wasted in early time steps.
- ➢ Core Causes
  - Gaussian prior carries zero information
    - → model must learn structure from scratch.
    - → curved, entangled velocity trajectory.
    - → required many sampling steps.
- ➢ Related Works
  - Existing work optimizes the trajectory, not the inefficient prior that causes it.

What is **RAPID**?

- ➢ Diagnoses sampling inefficiency at its source
  - → The prior, not the trajectory
- ➢ Structured GMM prior, principled integration into diffusion
- ➢ Coarse-to-fine design → high-quality few-step generation
- ➢ **RAPID** = **R**obust **A**daptive **P**rior **I**ntegration for **D**iffusion



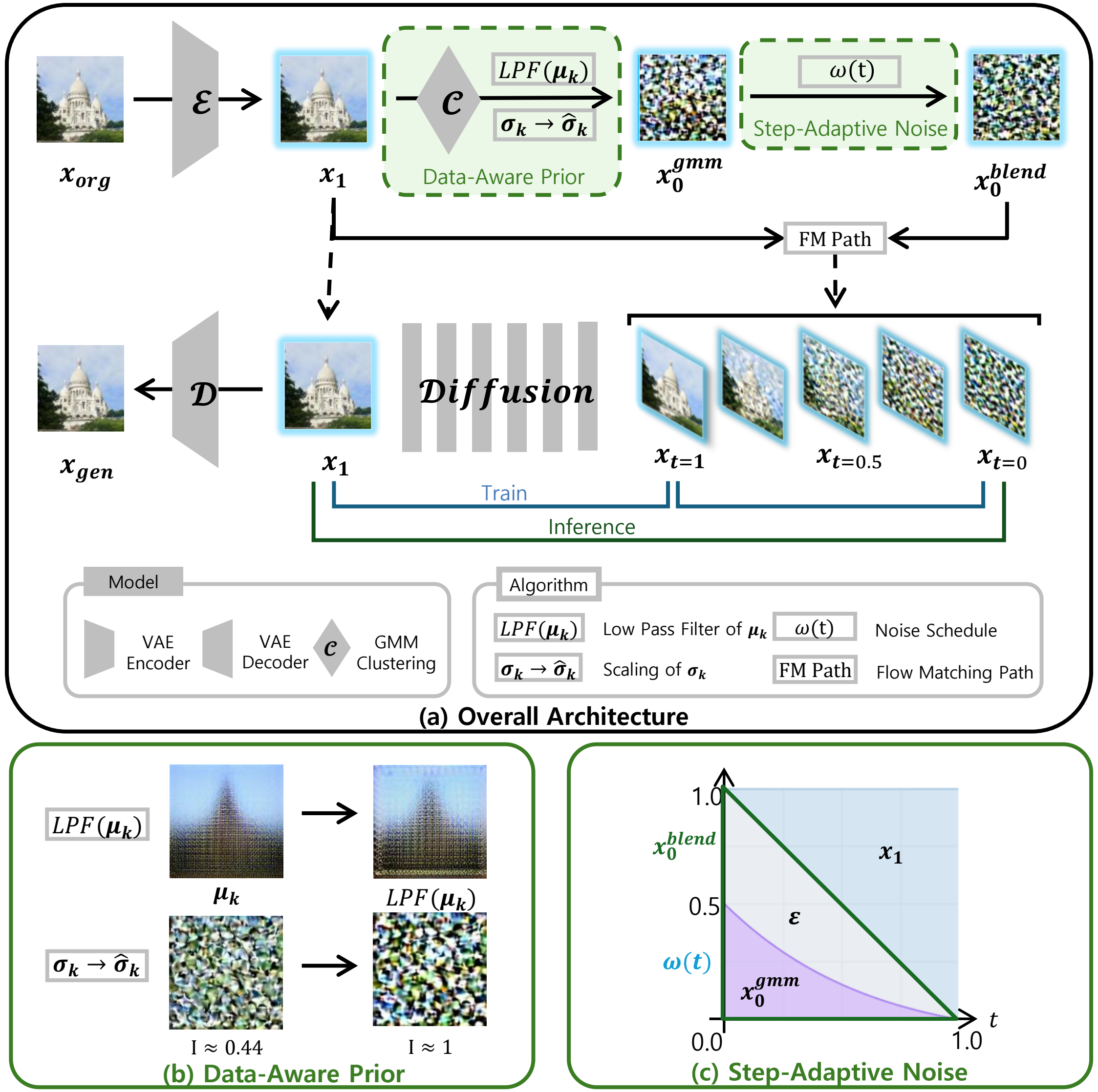
RAPID is 6.3x faster! **RAPID** is 84% fewer steps!

Master's Thesis (2/3)

Topic: Few-Step Diffusion via Adaptive Prior

Title: RAPID: Robust Adaptive Prior Integration Diffusion

Model Architecture



Method

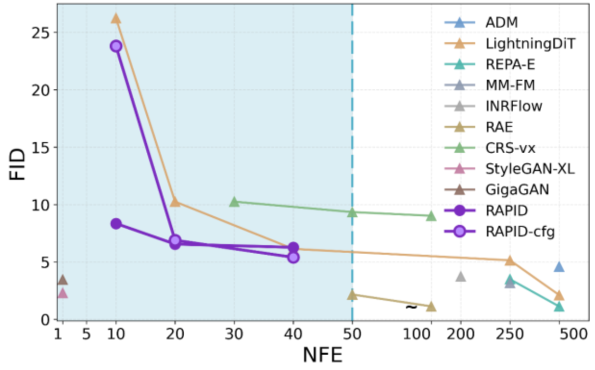
- Data-Aware Prior
  - GMM clustering of latent features
    - → reveals structural modes already in target dataset.
  - Cluster mean low-pass filter (LPF)
    - → keeps coarse structure only.
  - Cluster covariance normalization
    - → restores diversity for diffusion's stochastic.
- Step-Adaptive Noise
  - Coarse-to-fine principle
    - → GMM prior's role should fade over time.
  - SNR-aware blending w(t)
    - → controls this decay.
  - High w(t) early (structure) → low w(t) late (fine detail)

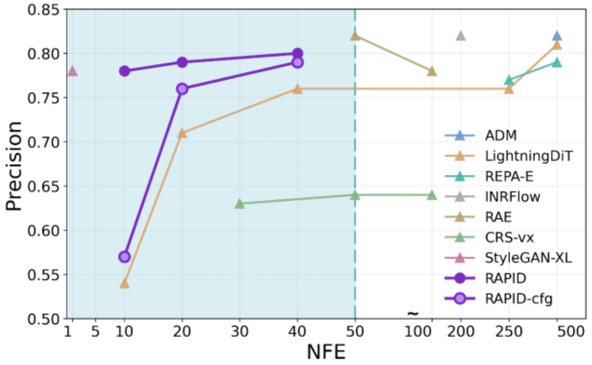
Master’s Thesis (3/3)

Topic: Few-Step Diffusion via Adaptive Prior

Title: RAPID: Robust Adaptive Prior Integration Diffusion

Experimental Results

- Quantitative Results
  - Strong performance preserved at 20 to 40 NFE(= few sampling step).
  - 40 NFE: FID 5.40 on ImageNet-1K, trained for 50 epochs
  - 



- Qualitativ

Achievement

- The adaptive prior approach underlying this thesis was developed through two preceding studies (First author).
  - "An Optimized Diffusion Model Based on Adaptive Prior Distribution," KMMS Autumn Conference, Nov. 2025. (Best Paper Award)
  - "Adaptive Prior Diffusion: Dataset-Aware GMM Prior for High-Fidelity Image Generation," KMMS Spring Conference, May 2026. (Best Paper Award)
- The current thesis work is being prepared for submission to *Pattern Recognition*.

Team Research Project I

Topic: Multimodal Empathetic Response Generation

Title: CAUSE: Cross-modal Attention for Utterance-level Speaker-listener

Research Objectives

- Multimodal (Text/Audio/Video) empathy generation.
- Cross-attention for Speaker-Listener emotional cues.
- Emotion-driven empathy modeling.

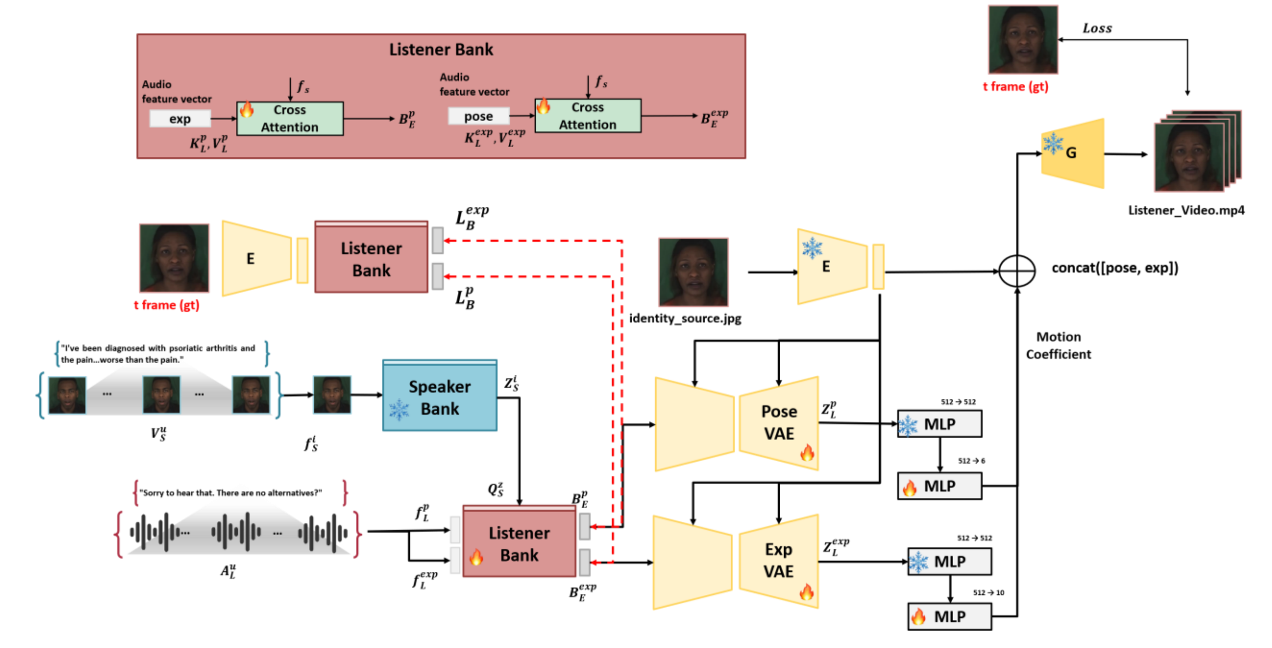
Motivation

- Problem Definition
  - Text-centric models in empathy research.
  - Ignoring non-verbal cues.
- Motivation
  - Varying empathy by facial/vocal tones.
  - Need for Speaker's multimodal fusion related to

Achievements

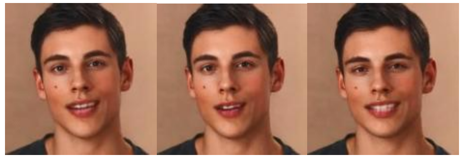
- The current thesis work is being prepared for submission to *AAAI 2027*.

Method

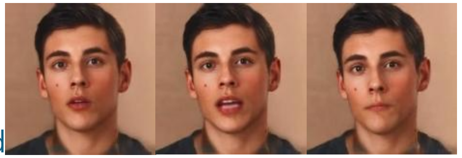


Experimental Results

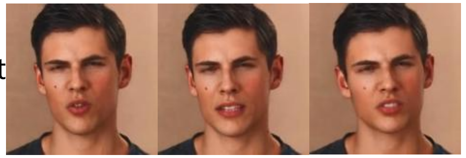
Speaker : happy Listener : happy



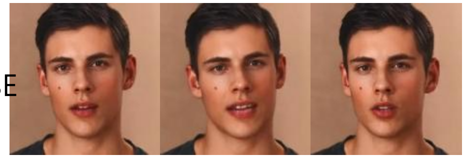
Speaker : sad Listener : surprised



Speaker : disgust Listener : angry



w/o CAUSE



Team Research Project Ⅱ

Research Objectives

- Multimodal emotion model via Dyadic datasets
- Vision & Audio integration for empathy.
- Multimodal alignment for consistency.

Motivation

- Problem Definition
  - Avatars limited to simple "talking head“.
  - Lack of high-dimensional cognitive empathy.
- Motivation
  - Learning empathy correlations in Dyadic interactions.

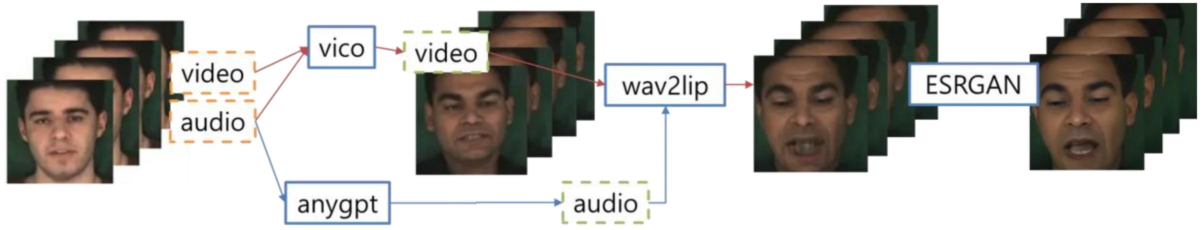
Achievements

- Published in *KMMS Autumn 2024* (First Author)
- Principal Investigator of “여대학(원)생 팀제연구지원사업 (한국여성과학기술인육성재단)”
- Principal Investigator of “숙명여대 SW중심대학”

Topic: Multimodal Emotion Recognition and Empathy Modeling

Title: Multi-Signal-Based User Emotion Recognition and Cognitive Empathy Modeling

Method



Vision & Audio integration

Multimodal alignment

Experimental Results

Speaker: neutral
Listener: happy

Speaker: neutral
Listener: worry

Speaker: neutral
Listener: furious

Personal Research

Topic: Transformer Attention Minification

Title: Attention Module Minification by Low-rank Factorization

Research Objectives

- ➢ Low-rank Factorization for Q, K, V matrices.
- ➢ Minimize FLOPs without accuracy loss.

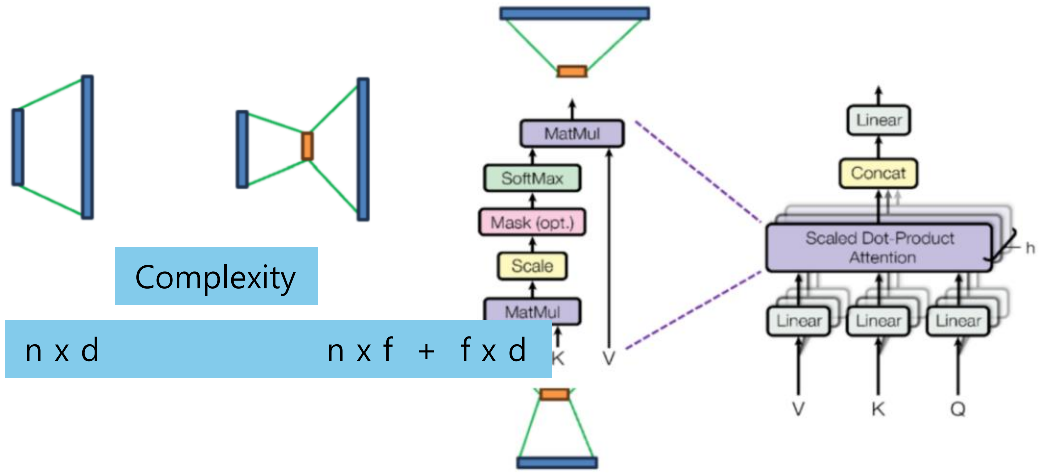
Motivation

- ➢ Problem Definition
  - ➢ Massive parameters and $O(n^2)$ complexity in matmul.
  - ➢ High computational overhead in Transformers.
- ➢ Motivation
  - ➢ Need for structural efficiency.
  - ➢ Maintaining performance with fewer calculations.

Achievements

- ➢ Published in *KMMS Spring 2025* (First Author & Best Paper Award).

Method



Experimental Results

- ➢ Minification
  - ➢ 14–25% vs baseline
  - ➢ 92.2–94.2% vs BERT
- ➢ Preserved performance with +0.004–0.008 accuracy improvement

Thank you for your attention.

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